

Critical Engagement with Sng (2014): A Structured Summary of Our Discussion

与Sng (2014) 的批判性对话：讨论结构化总结

How to read this document: Each point follows one of two structures. Where our discussion supplements Sng's argument, the structure is: **Sng's argument** → **Supplementary observation** → **Further historical and theoretical evidence**. Where our discussion challenges Sng's argument, the structure is: **Sng's argument** → **Challenging observation** → **Further historical and theoretical evidence**. Each section is marked accordingly.

阅读说明： 每一点遵循两种结构之一。当我们的讨论补充Sng的论点时，结构为：**Sng的观点** → **补充性观察** → **进一步的史实和理论证据**。当我们的讨论挑战Sng的论点时，结构为：**Sng的观点** → **挑战性观察** → **进一步的史实和理论证据**。每一节都相应标注。

Point 1: The Spatial Gradient of State Capacity — SUPPLEMENTARY

第一点：国家能力的空间梯度——补充

Sng's argument: China's geographic size created severe information asymmetries between the emperor in Beijing and his agents across the empire. Monitoring capacity declined with distance from the capital. Empirically, both land tax per capita and county density declined significantly with distance from Beijing, even after controlling for population density, elevation, and other factors. Every 1,000 km from Beijing was associated with a 25.9% lower land tax per capita and 25.7% fewer counties per 10,000 km² (pp. 114–118, Tables 3–4).

Sng的观点： 中国的地理规模在北京的皇帝与帝国各地代理人之间制造了严重的信息不对称。监督能力随距首都的距离递减。经验上，即使控制了人口密度、海拔等因素，人均土地税和县密度都随距北京的距离增加而显著下降。每距北京1000公里，人均土地税降低25.9%，每万平方公里的县数量减少25.7%（第114至118页，表3至4）。

Supplementary observation: Qin Hui's analysis of China's "big community standard" (大共同体本位) provides the institutional foundation for why this spatial gradient was so steep. From the Qin dynasty onward, the Legalist tradition systematically dissolved small communities — clans, village associations, guilds — that could have served as alternative channels of local governance and mutual monitoring. The state extended power directly to individual subjects ("编户齐民"), with no intermediate self-governing bodies to buffer or supplement the formal bureaucracy (Qin Hui, pp. 263, 269–271). In Western Europe, lords, parishes, and guilds partially performed governance functions at the local level, reducing the monarch's reliance on a single chain of bureaucratic agents. In China, because all small communities had been suppressed, the

emperor had no choice but to rely entirely on the formal bureaucratic chain. This made the monitoring problem that Sng identifies far worse than it would have been in a society with stronger local self-organization.

补充性观察： 秦晖对中国"大共同体本位"的分析，为这一空间梯度为何如此陡峭提供了制度基础。从秦朝开始，法家传统系统性地消解了小共同体——宗族、村社、行会——而这些组织本可以作为地方治理和互相监督的替代渠道。国家权力直接延伸到个体臣民（"编户齐民"），没有中间自治组织来缓冲或补充正式官僚体系（秦晖，第263、269至271页）。在西欧，领主、教区和行会在地方层面部分承担了治理功能，减少了君主对单一官僚代理人链条的依赖。在中国，由于所有小共同体被压制，皇帝别无选择只能完全依赖正式官僚链条。这使得Sng所识别的监督问题，比一个拥有更强地方自组织的社会要严重得多。

Further evidence: Qin Hui shows that in regions where small communities survived — the southeastern provinces of Guangdong, Zhejiang, and Jiangsu, where clan organizations were most active — resistance to collectivization in the 1950s was notably stronger than in the North China Plain, where atomized smallholders lacked any organizational basis for collective action (pp. 266–267). This confirms that the presence or absence of intermediate social organizations directly affected the state's ability to penetrate and control local society — a factor that Sng's model, focused solely on geographic distance, does not capture.

进一步证据： 秦晖表明，在小共同体存活的地带——宗族组织最活跃的东南沿海省份广东、浙江和江苏——1950年代对集体化的抵制明显强于华北平原，后者的原子化小农缺乏任何集体行动的组织基础（第266至267页）。这证实了中间社会组织的存在与否直接影响国家渗透和控制地方社会的能力——这是Sng的模型仅关注地理距离所未能捕捉的因素。

Point 2: "Tyranny at the Bottom" and the Sources of Peasant Suffering — SUPPLEMENTARY

第二点："底层暴政"与农民苦难的根源——补充

Sng's argument: Local agents exploited their informational advantage to extort taxpayers. The equilibrium extortion rate $\varepsilon^* = Y/(2mX)$ increases with the economy's size and decreases with monitoring capacity. The ruler keeps the official tax rate τ low because the total effective burden ($\tau + \varepsilon$) must not exceed the rebellion threshold r . Hence "high corruption and light taxation were causally linked" (p. 112).

Sng的观点： 地方代理人利用信息优势向纳税人勒索。均衡勒索率 $\varepsilon^* = Y/(2mX)$ 随经济规模增加而上升，随监督能力增强而下降。统治者将官方税率 τ 保持在低水平，因为总有效负担（ $\tau + \varepsilon$ ）不能超过叛乱门槛 r 。由此"高腐败与轻税负在因果上联系在一起"（第112页）。

Supplementary observation: Qin Hui's description of the "state broker" (国家经纪) role of village cadres provides a near-perfect empirical illustration of Sng's representative agent, extending the mechanism into the modern period. Post-reform administrative villages occupy an awkward

position: "serving the state without being supported by the state, supported by villagers without benefiting villagers" (p. 29). Their function, in peasants' words, is to "demand grain, demand money, demand lives" on behalf of higher authorities. When such agents are "neither officials nor commoners nor clan elders, yet manage land, people, and imperial grain" (p. 302), they tend to become a "negative interest stratum" that pursues its own enrichment — precisely the agent behavior Sng's model predicts.

补充性观察： 秦晖对村干部"国家经纪"角色的描述，为Sng的代表性代理人提供了近乎完美的经验例证，并将这一机制延伸到现代时期。改革后行政村处于尴尬位置："给国家办事而国家不养，由村民养活而无益于村"（第29页）。用农民的话说，其职能就是替上头"要粮要钱要命"。当这些代理人"非官非民非族长，管地管人管皇粮"（第302页）时，他们往往成为追求自身利益的"消极利益阶层"——正是Sng模型所预测的代理人行为。

Further evidence: Qin Hui's observation about the land reform movement is particularly striking. During Communist land reform, "the overwhelming majority of peasants' grievances were typically directed not at landlord exploitation through rents, but at the Nationalist government's crushing taxes, conscription, and the abuses of local baojia personnel" (p. 39). This directly supports Sng's claim that the primary source of peasant suffering in Chinese history was extraction by state agents and their proxies, not private landlord–tenant exploitation. It also supports Sng's argument that peasant revolts were triggered by violations of the no–revolt constraint ($\tau + \varepsilon > r$) rather than by private rent levels. Across all Chinese dynasties, Qin Hui argues, peasant wars were caused by "the court's abuse of corvée labor, excessive tax surcharges, severe economic controls that cut off the people's livelihoods, conflicts between strict household registration controls and waves of displaced people, or corrupt local administration" (pp. 44–45) — all about the state and its agents exceeding the population's tolerance threshold.

进一步证据： 秦晖关于土改运动的观察特别引人注目。在共产党土改中，"农民绝大部分的控诉通常并非冲着地租剥削，而是冲着国民党的苛捐杂税、拉丁派款和乡村保甲人员的为非作歹"（第39页）。这直接支持了Sng的论点：中国历史上农民苦难的主要来源是国家代理人及其代理的榨取，而非私人地主–佃农剥削。它还支持Sng的论点：农民叛乱由无反叛约束的违反 ($\tau + \varepsilon > r$) 而非私人地租水平所触发。秦晖论证，在所有中国朝代中，农民战争都是由"朝廷滥用民力、横征加派、严厉的经济统制绝民生路、官府严厉的户口控制与流民潮冲突、或由于吏治腐败"所引起的（第44至45页）——全都涉及国家及其代理人超越了民众忍耐阈值。

Point 3: Clerks Were Not Simply "Tyrants at the Bottom" — They Were a Complex, Double-Edged Institution — CHALLENGE

第三点：胥吏并非简单的"底层暴政"——他们是一个复杂的双刃制度——挑战

Sng's argument: In Sng's model, local agents are treated as pure rent–seekers who exploit information asymmetry to extort the population. The agent's only strategic choice is how much

to extract (the extortion rate ϵ), and the only constraint on this extraction is the probability of being caught and punished by the ruler. The agent has no social ties, no local identity, no interests other than personal enrichment. This one-dimensional characterization drives the entire model's predictions about corruption, low tax rates, and fiscal decline.

Sng的观点： 在Sng的模型中，地方代理人被视为纯粹的寻租者，利用信息不对称来勒索民众。代理人唯一的策略选择是榨取多少（勒索率 ϵ ），唯一的约束是被统治者发现和惩罚的概率。代理人没有社会联系、没有地方认同、没有个人牟利之外的利益。这种一维刻画驱动了模型关于腐败、低税率和财政衰退的全部预测。

Challenging observation: Historical evidence from the White Lotus movement reveals that clerks and runners were far more complex figures than Sng's model allows. They were not purely extractive agents of the state; they were embedded social actors whose local ties could make them conduits for anti-state mobilization — the exact opposite of their assigned role. During the Jia-Qing era White Lotus uprising, clerks in the Dangyang (当阳) religious network served as core organizers. They rallied co-religionists, coordinated with forces outside the city walls, and helped the rebels hold Dangyang city for five months. The very personnel who were supposed to be the state's frontline instruments for suppressing "heterodox sects" had become those sects' most effective organizers.

挑战性观察： 白莲教运动的历史证据揭示了胥吏远比Sng模型所允许的更为复杂。他们不是纯粹的国家榨取性代理人；他们是嵌入地方社会的行动者，其地方关系可以使他们成为反国家动员的渠道——这恰恰是其被赋予角色的反面。乾嘉之际白莲教起义中，当阳教团中的胥吏作为骨干，号召同教起事、接应城外同教，帮助叛军占据当阳城达五个月。本应作为国家镇压"邪教"最前线工具的人员，反而成了这些教派最有效的组织者。

Further evidence — the structural logic behind clerks' dual role:

进一步证据——胥吏双重角色背后的结构逻辑：

This was not accidental but structurally determined by several institutional features that Sng's model does not capture.

这并非偶然，而是由Sng模型未能捕捉的几个制度特征在结构上所决定的。

First, clerks occupied the position of "pivots between officials and the populace" (官民交接之枢纽) and bridges between city and countryside. Their daily work — collecting taxes, serving legal papers, conducting inspections, managing local registries — brought them into constant contact with all social strata. This gave them exactly the kind of dense, cross-cutting social network that the "big community standard" (大共同体本位) had destroyed among ordinary peasants. In Sng's model, agents are isolated utility-maximizers facing only the ruler above and taxpayers below. In reality, clerks were the most socially connected people in local society. Their occupational ties (业缘关系), kinship ties (血缘关系), and territorial ties (地缘关系) made them natural nodes for the spread of religious networks. The White Lotus teaching did not diffuse randomly; it spread along the pre-existing social channels that clerks' work had created —

including, critically, the inter-yamen communication networks that connected clerks in subordinate offices with those in superior offices, allowing the movement to break through geographic boundaries and achieve long-distance diffusion.

第一，胥吏处于"官民交接之枢纽"和城乡桥梁的位置。他们的日常工作——征税、送达法律文书、下乡巡查、管理地方簿册——使他们与各社会阶层保持持续接触。这赋予了他们恰恰是"大共同体本位"在普通农民中已经摧毁的那种密集的、交叉联系的社会网络。在Sng的模型中，代理人是孤立的效用最大化者，上面只面对统治者，下面只面对纳税人。在现实中，胥吏是地方社会中社会联系最广泛的人。他们的业缘关系、血缘关系和地缘关系使他们成为宗教网络传播的天然节点。白莲教的传播不是随机扩散的；它沿着胥吏工作所创造的既有社会渠道传播——关键的是，包括连接下级衙门胥吏与上级衙门胥吏的公务交往网络，使运动得以突破地域界限实现长距离扩散。

Second, the Qing deliberately rotated provincial-level officials frequently to prevent them from building local power bases. This "rule of avoidance" (回避制度) and frequent rotation meant that appointed officials almost never had deep knowledge of their jurisdictions' specific conditions. They therefore delegated most substantive administrative work to the permanently stationed clerks. This created a severe information asymmetry — but one that ran in the opposite direction from what Sng emphasizes. In Sng's model, the information problem is between the ruler and his agents: the ruler cannot observe the agent's extortion. In reality, there was an equally severe information problem between the rotating official and the permanent clerk: the official could not observe the clerk's actual activities, loyalties, or social connections. The clerk's stability and locality gave him an information monopoly that could be used not only for extortion (as Sng predicts) but also for protecting co-religionists, sheltering dissidents, filing false reports, and actively subverting the state's control functions. Whether a clerk actually wielded de facto power depended almost entirely on the personal quality of his superior — an institutional fragility far more specific than Sng's abstract "monitoring parameter" m .

第二，清朝刻意频繁调动省级官员以防止他们建立地方势力。这种"回避制度"和频繁调动意味着任命的官员几乎不可能对其辖区的具体情况有深入了解。他们因此将大多数实质性行政工作委托给长期驻留的胥吏。这创造了一种严重的信息不对称——但方向与Sng所强调的相反。在Sng的模型中，信息问题在于统治者与其代理人之间：统治者无法观察代理人的勒索。在现实中，流动的官员与常驻的胥吏之间存在同样严重的信息问题：官员无法观察胥吏的实际活动、忠诚或社会联系。胥吏的稳定性和地方性赋予他们信息垄断，这不仅可用于勒索（如Sng所预测），还可用于保护同教信徒、庇护异见者、呈报虚假信息、以及积极颠覆国家的控制职能。胥吏是否实际掌握权力几乎完全取决于其上司的个人素质——这是一种远比Sng抽象的"监督参数" m 更具体的制度脆弱性。

Third, the clerks' low status and miserable compensation — the very factor that Sng correctly identifies as driving extortion — also pushed them toward solidarity with the local population against the state. Deep-rooted local sentiment (乡土观念) made the state's formal mechanisms for suppressing heterodox sects — mutual responsibility systems (连坐) and informant incentives (举报) — virtually unenforceable, except in cases involving personal grudges. Clerks tasked with investigating and apprehending "heterodox sect members" had enormous discretionary power in defining who counted as a sect member and who did not. This

operational ambiguity explains why the White Lotus faith could endure across several dynasties of official suppression and even grow stronger under persecution: the very agents charged with eradicating it were also its protectors and propagators.

第三，胥吏低下的地位和微薄的报酬——Sng正确地将其识别为驱动勒索的因素——同时也推动他们与当地民众形成对抗国家的团结。深厚的乡土观念使国家镇压"邪教"的正式机制——连坐和举报——几乎无法有效执行，除非涉及个人恩怨。被指派缉查和追捕"邪教""教匪"的胥吏在界定谁算教徒、谁不算方面拥有巨大的自由裁量空间。这种操作上的模糊性解释了为什么白莲教信仰能绵延几个朝代的官方镇压而不断绝，反而在镇压下日益强盛：被赋予根除它的使命的代理人，同时也是它的保护者和传播者。

Fourth, and most importantly, some scholars have argued that the structural separation between "officials" and "clerks" (官吏分途) — the very institutional feature that created this dual role — was itself a rational response to the transaction costs of centralized governance. Keeping clerks as permanent, local, low-status functionaries did create the information asymmetry and corruption that Sng describes. But it also ensured continuity of local knowledge, maintained functioning administrative routines despite frequent official rotations, and provided a flexible (if exploitative) mechanism for adapting central directives to local conditions. This system was not purely dysfunctional; it was a second-best institutional solution to the genuine problem of governing a vast territory through a thin layer of rotating generalist officials.

第四，也是最重要的，一些学者认为"官吏分途"——正是创造了这种双重角色的制度特征——本身是对中央集权治理交易成本的理性应对。将胥吏保持为永久的、地方的、低地位的职能人员确实创造了Sng所描述的信息不对称和腐败。但它也确保了地方知识的连续性，在官员频繁轮换的情况下维持了行政运作的常规性，并提供了一种灵活的（虽然是剥削性的）机制，使中央指令能适应地方条件。这一制度并非纯粹功能失调；它是通过一薄层流动的通才官员治理广大领土这一真实问题的次优制度解决方案。

Implication for Sng's model: Sng treats the agent as a one-dimensional extractor. The White Lotus evidence shows that the agent was simultaneously an extractor, a local social broker, a potential rebel organizer, and an indispensable administrator. The "principal-agent problem" in late imperial China was not simply that agents stole too much; it was that the state had created agents who were deeply embedded in local society in ways that made them both necessary and uncontrollable. This is a fundamentally different kind of governance failure from the one Sng models, and it cannot be reduced to a distance-based monitoring problem. It is an institutional design problem — the specific way in which the Qing chose to structure the relationship between rotating officials and permanent clerks — that would exist regardless of the empire's geographic size.

对Sng模型的意涵： Sng将代理人视为一维的榨取者。白莲教的证据表明，代理人同时是榨取者、地方社会中介人、潜在的叛乱组织者和不可或缺的行政人员。晚期帝国中国的"委托-代理问题"不仅仅是代理人偷得太多；而是国家创造了深度嵌入地方社会的代理人，使他们既不可或缺又不可控制。这是一种与Sng所建模的根本不同的治理失败，它不能被简化为基于距离的监督问题。这是一个制度设计问题——清朝选择如何构建流动官员与常驻胥吏之间关系的具体方式——无论帝国的地理规模如何，这一问题都会存在。

Point 4: The Rebellion Threshold r May Have Been Endogenously High — CHALLENGE

第四点：叛乱门槛 r 可能是内生地很高——挑战

Sng's argument: The ruler sets the tax rate to ensure $\tau + \varepsilon \leq r$, where r is the rebellion threshold. This constraint is presented as relatively tight — the ruler must constantly worry about pushing the population past the breaking point (pp. 111–112).

Sng的观点： 统治者设定税率以确保 $\tau + \varepsilon \leq r$ ，其中 r 是叛乱门槛。这一约束被呈现为相对紧的——统治者必须不断担心把民众逼过承受极限（第111至112页）。

Challenging observation: Qin Hui's comparative analysis of collectivization in China and Russia suggests that r was endogenously much higher than Sng's model implies — precisely because the "big community standard" had destroyed the organizational capacity for collective resistance. In Russia, where autonomous village communes (mir) survived, peasants mounted fierce organized resistance to collectivization. In China, where atomized smallholders lacked organizational resources, collectivization "encountered no strong resistance and was accomplished faster than in Russia" (pp. 259–266), even though it represented a far more radical transformation. Qin Hui also documents that "heavy burdens, corrupt administration, and various disasters made peasants abandon their land and flee, even when they had land" (p. 51). The phrase "受田者与田为仇" ("those who received land became enemies of their land") captures a situation where the rational response to state extraction was not revolt but flight — suggesting a much higher tolerance threshold for collective action than Sng assumes.

挑战性观察： 秦晖对中俄集体化的比较分析表明， r 内生地远高于Sng模型所暗示的——正因为“大共同体本位”摧毁了集体抵抗的组织能力。在自治村社（米尔）存续的俄国，农民对集体化进行了激烈的有组织抵制。在原子化小农缺乏组织资源的中国，集体化“虽非农民所愿，却也未遇到强烈反抗便以比俄国更快的速度实现了”（第259至266页），尽管这是一个远更激进的转型。秦晖还记载，“沉重的负担、恶劣的吏治以及种种天灾人祸使农民有地不种、弃地而逃”（第51页）。“受田者与田为仇”捕捉了这样一种状况：面对国家榨取，理性反应不是反叛而是逃亡——暗示集体行动的忍耐门槛远高于Sng所假设的。

Further evidence and implication: If r was very high, then the reason for low taxes was not primarily fear of revolt (as Sng argues), but something else: institutional incapacity to collect more, or the political choice to maintain a low-cost governance model that preserved imperial power, or sheer administrative inertia rooted in "quota-ism" (原额主义). This does not invalidate Sng's model, but it suggests that the no-revolt constraint may have been less important than institutional constraints on the state's ability to expand its extractive capacity. Sng's model may be solving for the wrong binding constraint.

进一步证据及其意涵： 如果 r 非常高，那么低税收的原因主要不是对叛乱的恐惧（如Sng所论证），而是其他东西：收取更多税收的制度性无能，或者维持低成本治理模式以保持皇权的政治选择，或者根植

于"原额主义"的纯粹行政惯性。这并不否定Sng的模型，但暗示无反叛约束可能不如国家扩大汲取能力的制度性约束重要。Sng的模型可能在求解错误的约束条件。

Point 5: The "Paradox of Development" — SUPPLEMENTARY, with a CHALLENGE on Its Cause

第五点："发展悖论"——补充，但对其原因提出挑战

Sng's argument: Proposition 3 predicts that when agency costs are severe, economic expansion will initially increase but eventually depress tax revenue. The Qing's real tax revenue peaked in the early eighteenth century and then declined steadily despite continued economic growth. Revenue that could support 9.6% of the population in 1685 could support only 2.3% by 1848 (pp. 113, 119–120, Fig. 3).

Sng的观点：命题3预测当代理成本足够严重时，经济扩张最初将增加但最终会压低税收。清朝实际税收在十八世纪初达到峰值，此后尽管经济持续增长仍稳步下降。1685年税收能养活9.6%的人口，到1848年只能养活2.3%（第113、119至120页，图3）。

Supplementary observation: This matches well-known fiscal data: from the Qing's entry through the Pass (1644) to the eve of the Opium War, revenue increased by only about 169%, while population increased by roughly 400%. The hump-shaped pattern with peak under the Qianlong emperor followed by slow decline is a widely documented fact. Brook's *The Confusions of Pleasure* documents a strikingly similar process in the Ming dynasty: the central government "was retreating from tax administration, leaving it to local officials to figure out how to increase revenue" (p. 102), and officials were "forced to find ways to collect money on their own, which could not be formally recorded in the county's account books." The same dynamic of the formal fiscal system failing to keep pace with economic change is visible across multiple dynasties.

补充性观察：这与广为人知的财政数据吻合：从清朝入关（1644年）到鸦片战争前夕，岁入仅增加约169%，而人口增加了约400%。乾隆朝达到最高峰后缓慢下降的驼峰形模式是广泛记载的事实。卜正民的《纵乐的困惑》记录了明朝一个惊人相似的过程：中央政府"正退出税收的行政管理，交由地方官员去处理如何增加税收的问题"（第102页），官员们"被迫要自己去想办法征集金钱，而这又不能正式记录于县里的账簿里"。正式财政体系跟不上经济变迁的同一动态，在多个朝代都清晰可见。

Challenge on the cause: However, Sng attributes this "paradox" to the interaction between size and agency costs, treating the economy's growth as purely extensive ($Y = AN$, with constant productivity A). Brook's evidence shows that the economy's growth was not merely extensive. It involved qualitative structural changes: specialization of production stages in Jiangnan textile manufacturing, where "each stage of production became an independent productive activity" (p. 220); the shift from in-kind to silver-based taxation (the Single Whip Reform); and the development of long-distance commercial networks. These structural changes altered the

nature of the monitoring problem. Silver-based transactions are inherently more auditable than in-kind transactions — a shift to commercial taxation could have improved monitoring capacity without any change in geographic distance. The fact that the Ming and Qing states failed to exploit this opportunity suggests that the "paradox" was driven not by size making reform impossible, but by institutional rigidity making reform unattempted. The economy offered new possibilities for state revenue; the state's institutional apparatus was unable or unwilling to seize them.

对原因的挑战：然而，Sng将这一"悖论"归因于规模与代理成本的交互作用，将经济增长视为纯粹外延型的（ $Y = AN$ ，生产率 A 不变）。卜正民的证据表明经济增长不仅仅是外延型的。它涉及质的结构变化：江南纺织业生产阶段的专业化分工，"生产的每个环节都成为独立的生产活动"（第220页）；从实物税到白银税的转变（一条鞭法）；以及长距离商业网络的发展。这些结构性变化改变了监督问题的性质。白银交易本质上比实物交易更容易审计——向商业税的转变本可在不改变地理距离的情况下改善监督能力。明清国家未能利用这一机会表明，"悖论"不是因为规模使改革不可能，而是因为制度僵化使改革未被尝试。经济为国家收入提供了新的可能性；国家的制度机器无力或无意去抓住它们。

Point 6: Why the Fiscal System Did Not Budget for Actual Governance Needs — SUPPLEMENTARY, Deepening the Explanation

第六点：为什么财政体系不按实际治理需要编制预算——补充，深化解释

Sng's argument: A severe principal-agent problem leads to a small bureaucracy (Proposition 2). The equilibrium tax rate is too low to justify expanding the county system. In 1800, roughly 20,000 ranked officials governed over 300 million people (p. 113).

Sng的观点：严重的委托-代理问题导致小官僚机构（命题2）。均衡税率太低，无法证明扩展县制的合理性。1800年约20,000名有品级官员治理超过3亿人口（第113页）。

Supplementary observation, deepening the explanation: Sub-bureaucratic personnel (clerks, runners, private secretaries) who actually ran local government were deliberately kept in an informal, "off-budget" status. As Qu Tongzu's research documents, each county needed hundreds of these functionaries, but the formal budget provided for almost none of them. Runners received about 6 taels per year, far below subsistence. Their real income came from customary fees and extortion — which is the ε in Sng's model. This arrangement was not accidental. It served three functions simultaneously: (1) it saved central fiscal expenditure by shifting governance costs to local society; (2) it transferred the political risk of extraction — peasants blamed the clerks, not the emperor; (3) it kept agents in a state of permanent legal jeopardy (their income was technically illegal), giving the emperor perpetual grounds for punishment and thus maintaining asymmetric power. This was, from the emperor's perspective, not a "bad" institution but a "usable" one — until accumulated dysfunction made it unsustainable.

补充性观察，深化解释： 实际运转地方政府的编外人员（胥吏、衙役、幕友）被刻意保持在非正式的"编外"状态。如瞿同祖的研究所记载，每个县需要数百名这类人员才能运转，但正式预算几乎没有为他们提供经费。衙役年薪约6两，远低于维生所需。他们的实际收入来自惯例收费和勒索——这就是Sng模型中的 ϵ 。这种安排并非偶然。它同时服务于三重功能：（1）将治理成本从中央财库转嫁给地方社会，节省中央财政支出；（2）转移了榨取的政治风险——农民怨恨胥吏而非皇帝；（3）使代理人处于永久的法律不安全状态（其收入在技术上是非法的），给予皇帝永久的惩罚理由，从而维持不对称的权力关系。从皇帝的视角看，这不是"坏"制度而是"可用"的制度——直到积累的功能失调使其不可持续。

Further evidence: Zelin's *The Magistrate's Tael* shows that Yongzheng's "nourish honesty silver" (养廉银) reforms raised stipends roughly 30-fold but did not eliminate corruption, because as the economy grew, rent-seeking temptations grew proportionally. Sng cites Ni and Van's (2006) estimate that eliminating corruption in 1750 would have required a 250-fold wage increase (p. 123). But beyond this "efficiency wage" problem, Qin Hui identifies a deeper political-structural obstacle: without taxpayer rights — what he calls an "identity-based tribute" (身份性贡赋) system rather than a rights-and-obligations-based tax system — there was no institutional mechanism for a "tax-for-services" bargain (p. 25). In European parliamentary traditions, taxpayers consented to higher taxes in exchange for specific public services, creating a positive feedback loop between taxation and governance quality. In China, taxation was a unilateral obligation with no corresponding entitlement to services, so no constituency demanded that the state professionalize or expand its bureaucracy — even when doing so would have benefited the population.

进一步证据： 曾小萍（Zelin）的《州县官的银两》表明雍正的"养廉银"改革将俸禄提高了约30倍，但未能消除腐败，因为随着经济增长，寻租诱惑也等比增长。Sng引用倪志伟和Van（2006）的估算，认为在1750年消除腐败需要250倍的工资增长（第123页）。但超越这一"效率工资"问题，秦晖识别了更深层的政治-结构性障碍：在没有纳税人权利的情况下——他称之为"身份性贡赋"体制而非"权利-义务统一"的税制——不存在"以税换服务"的制度性机制（第25页）。在欧洲议会传统中，纳税人同意更高税收以换取特定公共服务，在税收与治理质量之间创造了正反馈循环。在中国，纳税是单方面义务，没有相应的获得服务的权利，因此没有利益群体要求国家专业化或扩大其官僚机构——即使这样做会让民众受益。

Point 7: Why Commercial Development Did Not Lead to Political Reform or Fiscal Expansion — CHALLENGE

第七点：为什么商业发展没有导向政治改革或财政扩张——挑战

Sng's argument: Sng does not directly address this question. His model treats the economy as an exogenous variable and focuses on how its growth affects fiscal capacity through the principal-agent channel. He notes that China's peaceful external environment reduced the

pressure to build fiscal capacity (p. 119), but does not explore why the thriving commercial economy did not generate pressure for institutional change from within.

Sng的观点： Sng没有直接讨论这个问题。他的模型将经济视为外生变量，聚焦于经济增长如何通过委托-代理渠道影响财政能力。他指出中国和平的外部环境降低了建设财政能力的压力（第119页），但没有探讨为什么繁荣的商业经济没有从内部产生制度变革的压力。

Challenging observation: This is a major gap in Sng's analysis. The supplementary literature reveals several interconnected reasons why commercial growth failed to translate into political reform or fiscal modernization — none of which are captured by a "size" variable.

挑战性观察： 这是Sng分析的一个重大缺口。补充文献揭示了商业增长未能转化为政治改革或财政现代化的几个相互关联的原因——没有一个能被"规模"变量所捕捉。

Further evidence — four mechanisms:

进一步证据——四重机制：

(1) Merchants were absorbed into the gentry system rather than forming an independent political class. Brook demonstrates that the social distance between merchants and gentry "was constantly being narrowed, and this narrowing strengthened rather than weakened the gentry social group" (p. 291). Merchants sought gentry status through land purchase, cultural investment, and marriage alliances. They did not seek to build alternative political institutions. Brook explicitly invokes Braudel: China developed "an extensive market economy... but this was not capitalism in the European sense" (p. 227). The key difference was the social structure within which commerce operated.

(1) 商人被吸纳进绅士体系而非形成独立政治阶层。 卜正民证明商人与绅士之间的社会距离"不断被拉近，而这种弥合的方式加强而不是削弱了士绅社会集团的力量"（第291页）。商人通过购地、文化投资和联姻追求绅士身份，而不寻求建立替代性政治制度。卜正民明确引用布罗代尔：中国发展了"一个广泛的市场经济……但这不是欧洲意义上的资本主义"（第227页）。关键差异在于商业运作于其中的社会结构。

(2) State monopoly over key revenue sources created patron–client relations, not political bargaining. Merchants in the most profitable sectors (salt, tea, mining) operated under state licenses and depended on state permission for their profits. As Brook documents regarding Huizhou salt merchants, their strategy was to "find opportunities to enter officials' cultural circles" (pp. 144–145) to reduce official predation and gain competitive advantages — the opposite of the European pattern where independent merchant wealth enabled demands for political concessions.

(2) 国家对关键税源的垄断创造了庇护-依附关系而非政治谈判。 最有利可图行业（盐、茶、矿业）的商人在国家许可下经营，利润依赖国家准许。正如卜正民记述的徽州盐商，他们的策略是"努力寻找机会进入官员的文化圈子"（第144至145页）以减少官府掠夺和获取竞争优势——这与欧洲模式正相反，在欧洲，独立的商人财富使他们能够要求政治让步。

(3) No organizational infrastructure for collective merchant political action. Qin Hui's "big community standard" framework explains this directly: independent social organizations were systematically suppressed. Merchants had no equivalent of European urban communes, chartered guilds, or representative assemblies. As Liang Shuming observed (quoted in Qin Hui, p. 271): "For a Chinese person, what is close at hand is the family; what is distant is all-under-heaven... the middle ground suitable for cultivating group life" was missing. Without intermediate organizations, merchants could only interact with the state as individuals, never as a collective political force.

(3) 商人缺乏集体政治行动的组织基础设施。 秦晖的"大共同体本位"框架直接解释了这一点：独立的社会组织被系统性地压制。商人没有欧洲城市公社、拥有特许权的行会或代议机构的对等物。如梁漱溟所观察（转引自秦晖，第271页）："中国人切己的便是身家，远大的便是天下了……得乎其中，有一适当范围，正好培养团体生活"的中间地带是缺失的。没有中间组织，商人只能以个人身份与国家互动，而永远无法作为集体政治力量。

(4) Policy alternatives existed but were not adopted — indicating political, not geographic, obstacles. Brook documents that Qiu Jun's (丘濬) *Supplement to the Great Learning* (1487) argued against state monopolies, for free maritime trade, and for letting merchants manage production and distribution (pp. 116–117). The Ming intellectual tradition knew what more market-friendly policies would look like. The fact that such ideas were never implemented points to the political logic of a system that would not empower social groups outside the existing hierarchy.

(4) 政策替代方案存在但未被采纳——表明障碍是政治性的而非地理性的。 卜正民记载丘濬的《大学衍义补》（1487年）反对国家垄断，支持自由海上贸易，主张让商人管理生产和分配（第116至117页）。明代知识传统知道更友好的市场政策长什么样子。这些思想从未被实施，指向的是一个不愿赋权于现有等级体系之外社会群体的制度的政治逻辑。

Point 8: "Without Foreign Invasion, the Qing Might Have Fallen Sooner" — CHALLENGE

第八点："没有洋人入侵，清朝可能灭亡更快"——挑战

Sng's argument: Sng treats the Opium War and subsequent foreign invasions primarily as external shocks that exposed the Qing state's accumulated weakness, especially its underinvestment in military capacity due to fiscal constraints (pp. 119–123).

Sng的观点： Sng将鸦片战争及此后的外国入侵主要视为暴露清朝积累的弱点的外部冲击，尤其是因财政约束导致的军事投入不足（第119至123页）。

Challenging observation: The causal relationship may have been more complex — indeed, partially reversed. Without the new revenue sources created by forced opening (customs duties administered by the Imperial Maritime Customs Service under Robert Hart, and the likin 厘金 tax

on internal commerce introduced during the Taiping Rebellion), the Qing likely could not have survived the mid–nineteenth–century internal crises. Before the Opium War, prior dynasties almost all fell to large–scale peasant revolts. The Qing was already facing the same fiscal exhaustion that had destroyed previous dynasties — per capita revenue had been declining for over a century. The foreign–managed customs service and the new commercial taxes broke through a fiscal ceiling that the traditional land–tax system could never have breached.

挑战性观察： 因果关系可能更复杂——实际上部分是反过来的。如果没有被迫开放所创造的新税源（赫德主持的海关总税务司所征收的关税，以及太平天国时期引入的厘金），清朝很可能熬不过十九世纪中期的内部危机。鸦片战争之前，此前各朝几乎都灭亡于大规模农民起义。清朝已经面临着摧毁此前各朝的同样的财政枯竭——人均税收已经持续下降了一个多世纪。外国人管理的海关和新的商业税收突破了传统土地税制永远无法突破的财政天花板。

Further evidence: Late Qing fiscal data dramatically illustrates this point. Revenue exceeded 100 million taels in 1899, crossed 200 million in 1903, and reached nearly 300 million by 1911 — an explosion of fiscal capacity unimaginable in a purely agricultural tax regime. This was concrete evidence that Chinese society was already undergoing a transformation toward a commercial–industrial economy. The critical point is: this revenue explosion occurred on the same territory with essentially the same population. Size did not change. What changed was the institutional channel of extraction — customs duties, commercial transit taxes, and eventually modern–style taxes on trade and manufacturing. If size were the binding constraint (as Sng argues), then new institutional channels should not have made such a dramatic difference.

进一步证据： 晚清财政数据戏剧性地说明了这一点。1899年岁入超过1亿两，1903年超过2亿两，1911年将近3亿两——这是纯农业税制下不可想象的财政能力爆发。这是中国社会正在转型为工商业社会的具体证据。关键在于：这一收入爆发发生在同一片领土上，人口本质上相同。规模没有改变。改变的是汲取的制度渠道——关税、商业过境税，以及最终的现代型贸易和制造业税收。如果规模是约束性瓶颈（如Sng所论证），那么新的制度渠道不应该造成如此戏剧性的差异。

Moreover, the Maritime Customs Service itself refutes the size thesis. It operated across the entire Chinese coastline — a vast geographic span — yet achieved levels of fiscal efficiency that the traditional bureaucracy never approached. This was not because the coastline was shorter than the interior, but because the institutional design was fundamentally different: professional staff, standardized procedures, independent auditing, and vertical management insulated from the traditional bureaucratic hierarchy.

更有甚者，海关总税务司本身就反驳了规模论。它运作于整个中国海岸线——一个巨大的地理跨度——却达到了传统官僚体系从未接近的财政效率。这不是因为海岸线比内陆短，而是因为制度设计根本不同：专业人员、标准化程序、独立审计和不受传统官僚层级干扰的垂直管理。

Point 9: The KMT's Fiscal Structure — Extending the Institutional Argument Beyond the Qing

第九点：国民党的财政结构——将制度论证延伸到清朝之后

Sng's argument: Sng's analysis ends at 1850 and does not discuss subsequent fiscal developments.

Sng的观点： Sng的分析止于1850年，未讨论此后的财政发展。

Extending the analysis: The Nationalist (KMT) government's fiscal structure provides a further test case that confirms the primacy of institutional design over geographic size. The KMT derived the majority of its revenue from customs duties and coastal salt taxes — taxes that did not require deep engagement with rural society. At its peak in 1929, customs revenue constituted 51% of total government revenue. The KMT was essentially a regime that "taxed foreigners" through import duties. Coastal salt taxes added another 10% or more. This meant the state had virtually no institutional linkage with the vast interior agricultural population — it neither taxed them significantly nor invested in them. The fiscal base depended on the volume of imports and coastal trade, not on the productivity or welfare of the rural majority.

延伸分析： 国民党政府的财政结构提供了一个进一步的检验案例，确认了制度设计优先于地理规模。国民党的大部分收入来自关税和沿海盐税——这些税种不需要与农村社会深度接触。1929年高峰期，关税收入占政府总收入的51%。国民党本质上是一个通过进口关税"向外国人收税"的政权。沿海盐税又占10%以上。这意味着国家与广大内陆农业人口之间几乎没有制度性联系——既不大量向他们征税，也不向他们投资。财政基础依赖进口量和沿海贸易，而非农村多数人口的生产力或福祉。

This fiscal structure had two consequences directly relevant to Sng's framework. First, it made the KMT structurally incapable of deep reform: customs and salt revenues were largely pledged to repay foreign debts and cover military expenditures, leaving little for productive investment or rural administration. Second, it put the KMT's interests in direct conflict with domestic capitalists — the regime depended on imports, while domestic industry would have benefited from protection. The result was a government fiscally disconnected from the society it nominally governed, unable to mobilize resources for rural development, land reform, or grassroots administrative capacity. This was not a consequence of China's size. It was a consequence of the state's institutional choices about how and from whom to extract revenue.

这种财政结构对Sng的框架有两个直接相关的后果。第一，它使国民党在结构上无力进行深层改革：关税和盐税收入大部分被用于偿还外债和军费开支，几乎没有余力用于生产性投资或农村行政。第二，它使国民党的利益与国内资本家直接冲突——政权依赖进口，而国内工业本将受益于保护政策。结果是一个在财政上与其名义治理的社会脱节的政府，无力动员资源用于农村发展、土地改革或基层行政能力建设。这不是中国规模的后果。这是国家关于如何以及向谁汲取收入的制度选择的后果。

The Communist Party, governing the same territory, chose a fundamentally different approach — penetrating rural society through party organization and extracting surplus through collective

institutions — and achieved dramatically different results, confirming that the binding constraint was institutional, not geographic.

共产党治理着同一片领土，选择了根本不同的方式——通过党组织渗透农村社会、通过集体制度汲取剩余——并取得了戏剧性不同的结果，确认了约束性瓶颈是制度性的，而非地理性的。

Point 10: The Fundamental Problem Was Institutional Obsolescence, Not Size — MAJOR CHALLENGE

第十点：根本问题是制度过时，而非规模——重大挑战

Sng's argument: Size is the key independent variable. Geographic size determines the monitoring parameter m , which in turn determines corruption, tax rates, bureaucratic size, and the dynamic trajectory of fiscal capacity. The parameter m did not improve significantly during the late imperial period due to the absence of improvements in information technology (p. 122).

Sng的观点： 规模是关键自变量。地理规模决定监督参数 m ， m 进而决定腐败、税率、官僚规模和财政能力的动态轨迹。由于信息技术没有显著改善，参数 m 在晚期帝国时期没有明显提高（第122页）。

Major challenge: The strongest counterargument to Sng's size thesis comes from the modern Chinese experience. The People's Republic governs essentially the same territory with dramatically greater state capacity. This was achieved not by shrinking the territory but by fundamentally redesigning the institutional "operating system." The Communist Party raised m through organizational innovation, not through geographic change: (1) Party cells penetrated to the village level, creating a parallel monitoring hierarchy the imperial system never had. (2) Land reform destroyed the local gentry class — Sng's "representative agent" — and replaced it with cadres whose careers depended on the party hierarchy. (3) Collectivization and the unified purchase and sale system (统购统销) allowed the state to directly appropriate agricultural surplus without relying on local intermediaries. (4) The 1994 tax-sharing reform (分税制) restructured central-local fiscal relations, and modern information systems dramatically improved monitoring capacity. These were institutional innovations, not geographic changes.

重大挑战： 对Sng规模论的最强反驳来自现代中国的经验。中华人民共和国治理着本质上相同的领土，但国家能力戏剧性增强。这不是通过缩小领土实现的，而是通过根本性地重新设计制度"操作系统"。共产党通过组织创新而非地理变化提高了 m ：(1) 党支部渗透到村一级，创建了帝国制度从未有过的平行监督层级。(2) 土改摧毁了地方绅士阶层——Sng的"代表性代理人"——代之以职业生涯依赖党的层级的干部。(3) 集体化和统购统销制度使国家能直接占有农业剩余，不依赖地方中间人。(4) 1994年分税制改革重构了中央-地方财政关系，现代信息系统戏剧性地改善了监督能力。这些是制度创新，而非地理变化。

This suggests that Sng's model confounds two distinct variables: geographic size (which is genuinely exogenous and constant) and institutional design (which is endogenous and changeable). The late imperial state was weak not because China was large, but because its

institutional "software" had become obsolete relative to the society it was governing. The Chinese bureaucratic system, though it appeared advanced and matured early, had long since ceased to be adequate to the economy's actual form. Its officials' skills, specialization, and knowledge were designed for a simple agrarian society, not for the complex commercial economy that had developed by the Ming–Qing period. It could persist as long as there were no external military threats; the Opium War exposed the mismatch.

这表明Sng的模型混淆了两个不同的变量：地理规模（真正外生且恒定）和制度设计（内生且可变）。晚期帝国国家的虚弱不是因为中国大，而是因为其制度"软件"相对于它治理的社会已经过时。中国的官僚制度虽然看起来很先进且成熟很早，但其官员的技能、专业化和知识早已不适应经济发展的实际形态。它是为简单的农业社会而设计的，而不是为明清时期已经发展起来的复杂商业经济而设计的。它之所以能一直维持，是因为没有外部军事威胁；鸦片战争暴露了这种错配。

Further evidence — five scholarly traditions support this institutional–obsolescence diagnosis over the size thesis:

进一步证据——五个学术传统支持制度过时的诊断而非规模论：

(1) Ray Huang's "management by numbers" (数目字管理) thesis. The traditional state could not precisely measure, record, or audit economic activity — not because of distance, but because the entire administrative system was not designed for precision management. Tax quotas were set by ancestral precedent, not by actual economic output. Financial records were so chaotic that even the Board of Revenue could not truly verify provincial reports.

(1) 黄仁宇的"数目字管理"说。 传统国家无法精确地测量、记录和审计经济活动——不是因为距离远，而是因为整个行政体系的设计理念就不以精确管理为目标。税额按祖制设定，而非按实际经济产出动态调整。财务记录混乱到连户部都无法真正核实各省上报的数字。

(2) Iwai Shigeki's "quota-ism" (原額主義) concept. The Ming–Qing fiscal system's defining feature was fixing revenues and expenditures at permanent quotas, rarely adjusted once set. This guaranteed stability but eliminated all flexibility. When the economy grew, population shifted, or new governance demands appeared, the quota remained unchanged, and the state's real fiscal capacity shrank in relative terms. This was an institutional design problem, not a distance problem.

(2) 岩井茂树的"原额主义"概念。 明清财政制度的根本特征是将各项收支固定为"定额"，一旦确定就极少调整。这保证了稳定性但消除了所有灵活性。当经济增长、人口转移或新治理需求出现时，定额不变，国家的实际财政能力在相对意义上萎缩。这是制度设计问题，不是距离问题。

(3) Wang Yeh–chien's (王业键) fiscal history of the Qing. Wang documents that the Qing tax system had no income tax concept, no property assessment system, and no periodic land surveys or tax–base updates. "Yongbu jiashui" ("永不加赋," "taxes will never be increased") was not just a political promise but reflected an institutional reality: the state simply did not possess the administrative capacity to assess how much to increase taxes or from whom. This is a capacity deficit, not a size deficit.

(3) 王业键的清代财政史研究。王业键的研究表明，清代税制没有所得税概念，没有财产评估制度，没有定期的土地普查和税基更新。“永不加赋”不只是政治承诺，它反映了制度现实：国家根本不具备评估应该向谁加多少税的行政能力。这是能力缺陷，不是规模缺陷。

(4) Philip Kuhn's study of the 1768 "soulstealers" scare (叫魂案). Kuhn (cited by Sng, p. 116) shows that even the most energetic emperor (Qianlong), confronting a nationwide incident, could not distinguish real information from bureaucratic obfuscation. This is indeed a monitoring problem, but Kuhn's analysis reveals it was institutional, not merely geographic: the Qing had no information-gathering apparatus independent of the bureaucracy itself. No independent statistical office, no independent auditing body — the emperor could only set bureaucrats to watch bureaucrats, and the watchers could be co-opted.

(4) 孔飞力对1768年“叫魂”案的研究。孔飞力（Sng引用，第116页）展示了即使是最精力充沛的皇帝（乾隆），面对一个全国性事件，也无法区分真实信息和官僚的敷衍塞责。这确实是监督问题，但孔飞力的分析揭示它是制度性的，而非仅仅地理性的：清朝没有独立于官僚体系本身的信息收集机构。没有独立的统计机构，没有独立的审计机构——皇帝只能设置官僚监督官僚，而监督者也可能被收买。

(5) The civil examination system as an elite-absorption mechanism. The examination system channeled elite ambition into the existing bureaucratic structure. Wealthy merchants' rational strategy was to invest in examination preparation and cultural capital to gain gentry status (Brook, pp. 242, 291), not to build autonomous political organizations. This prevented the emergence of an independent bourgeoisie that might have pressured the state to modernize its fiscal and administrative systems — the trajectory that occurred in parts of Western Europe.

(5) 科举制度作为精英吸纳机制。科举制度将精英抱负引导进现有官僚结构。富裕商人的理性策略是投资于科举准备和文化资本以获取绅士身份（卜正民，第242、291页），而非建立自主的政治组织。这阻止了独立资产阶级的出现——独立资产阶级本可能向国家施压，要求其财政和行政体系现代化——这正是西欧部分地区曾出现的轨迹。

Point 11: The Broader Pattern — States Outgrow Their Institutional Capacity — SUPPLEMENTARY, Reframed

第十一点：更广泛的模式——国家超越其制度能力——补充，重新框定

Sng's argument: The "paradox of development" — economic growth weakening state fiscal capacity when agency costs are severe — is specific to the principal-agent dynamics of large, centralized, weakly monitored states (pp. 108, 113, 119).

Sng的观点：“发展悖论”——当代理成本严重时经济增长削弱国家财政能力——是大型、中央集权、监督薄弱的国家特有的委托-代理动态（第108、113、119页）。

Reframing: Japanese material from Kang Hao's study suggests that a similar pattern — the state's responsibilities outgrowing its institutional capacity — can occur even in small polities.

The Kamakura Bakufu, governing a far smaller territory, experienced a version of the same trap after the Mongol invasions: it acquired unprecedented responsibilities but "many of these matters were originally outside the Bakufu's jurisdiction; after the Mongol invasions, the Bakufu became the outlet for all contradictions" (Kang Hao, p. 83). The result was institutional overstretch and collapse — not because Japan was large, but because institutional capacity could not keep pace with demands.

重新框定： 康昊研究中的日本材料表明，类似的模式——国家职责超越其制度能力——即使在小政体中也可能发生。镰仓幕府治理着远更小的领土，在蒙古入侵后经历了同一陷阱的一个版本：它获得了前所未有的职责，但"很多事情原本并不在幕府的职权范围之内，蒙古袭来以后，幕府变成了一切矛盾的出口"（康昊，第83页）。结果是制度过度扩张和崩溃——不是因为日本大，而是因为制度能力跟不上需求。

Further evidence and general principle: This suggests a reframing of Sng's insight. The "paradox of development" is not specifically about size but about the gap between institutional capacity and governance demands. Premodern states faced rigid constraints: narrow tax base definitions (mainly land), high fixed costs of administrative expansion (adding a county meant an entire apparatus), and no credit instruments (no bonds, no viable paper money — the Ming tried and failed). Modern states can respond elastically through debt issuance, monetary policy, and adjustable tax types. The critical question is not "how big is the territory?" but "can the institutional system upgrade itself in response to new demands?" Size amplifies the gap (larger territories generate more diverse demands), but the rigidity of institutional design determines whether the gap becomes fatal. China's size made the problem more acute; it did not create it.

进一步证据和一般性原理： 这建议对Sng的洞见进行重新框定。"发展悖论"并非特别关于规模，而是关于制度能力与治理需求之间的差距。前现代国家面临刚性约束：狭窄的税基定义（主要是土地）、行政扩张的高固定成本（增设一县就意味着整套机构）、以及缺乏信用工具（没有债券，没有可行的纸币——明朝尝试过但失败了）。现代国家可以通过发行债务、货币政策和可调节的税种进行弹性回应。关键问题不是"领土有多大？"而是"制度体系能否自我升级以回应新的需求？"规模放大了差距（更大领土产生更多样的需求），但制度设计的僵化决定了差距是否变得致命。中国的规模使问题更加尖锐；但并非创造了它。

Point 12: A Fair Assessment of Sng's Contribution

第十二点：对Sng贡献的公允评估

Sng's paper remains a significant contribution. His model captures a genuine mechanism — the principal-agent problem in hierarchical bureaucracies is real, and geographic distance does exacerbate it. His empirical contribution — the spatial gradient of taxation and administration measured with original data, the reconstructed fiscal time series — is valuable. The "paradox of development" is a genuinely novel theoretical proposition. What this discussion suggests is not that Sng is wrong, but that his explanation is incomplete.

Sng的论文仍然是一项重要贡献。他的模型捕捉了一个真实的机制——层级官僚体系中的委托-代理问题是真实的，地理距离确实加剧了它。他的经验贡献——用原创数据测量的税收和行政的空间梯度、重建的财政时间序列——是有价值的。"发展悖论"是一个真正新颖的理论命题。本讨论所表明的不是Sng是错误的，而是他的解释是不完整的。

More precisely, Sng confounds two distinct variables: geographic size (genuinely exogenous and constant) and institutional design (endogenous and changeable). The late imperial Chinese state's fiscal weakness was the joint product of a challenging geography and an institutional apparatus that had failed to evolve for centuries. The same territory can be governed with vastly different levels of effectiveness depending on the institutional "software" deployed — as the dramatic contrasts between the Qing, the KMT, and the PRC demonstrate. The traditional Chinese bureaucratic system matured early and appeared advanced, but its officials' skills, knowledge, and organizational methods had long since ceased to match the economy's actual complexity. It could persist without external military pressure; the Opium War exposed the mismatch. The root of dynastic decline was not that China was too large to govern, but that its governing institutions had stopped developing while its economy and society had not.

更准确地说，Sng混淆了两个不同的变量：地理规模（真正外生且恒定）和制度设计（内生且可变）。晚期中华帝国国家的财政虚弱是具有挑战性的地理与几个世纪未能进化的制度机器的共同产物。同一片领土可以根据部署的制度"软件"以截然不同的效力水平来治理——正如清朝、国民党和中华人民共和国之间的戏剧性对比所展示的。中国传统的官僚制度虽然成熟很早、看起来很先进，但其官员的技能、知识和组织方式早已不再匹配经济的实际复杂性。在没有外部军事压力的情况下它可以持续存在；鸦片战争暴露了这种错配。王朝衰落的根源不是中国太大而无法治理，而是其治理制度停止了发展，而其经济和社会却没有。